

Code-Layout, Readability and Reusability

Date	04 july 2026
Team ID	XXXXXX
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase for the Human Development Index (HDI) Prediction Using Machine Learning project. The project follows a modular and organized coding approach, making it easy to understand, maintain, and extend. Proper naming conventions, comments, reusable functions, and structured folders improve the overall quality of the application.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation	Yes	Improves readability
2	Proper File Structure	Files and folders are logically	Yes	Separate folders for templates and
3	Meaningful Variable Names	Variables clearly represent their	Yes	Easy to understand
4	Function / Method Names	Functions use descriptive	Yes	Simple and meaningful
5	Code Comments	Important sections are	Yes	Helpful for maintenance
6	Modular Design	Code divided into reusable	Yes	Training and Flask code separated
7	No Redundant Code	Duplicate or unused code	Yes	Optimized implementation
8	Error Handling	Handles invalid inputs and	Partial	Basic input validation

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	app.py	Python (Flask)	Web application backend	High
2	train_model.py	Python	Model training	High
3	model.pkl	Pickle	Prediction module	High
4	index.html	HTML/CSS	User Interface	High
5	Data Preprocessing	Python (Pandas)	Model training	Medium

Overall Code Quality Assessment

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well-organized folder structure and modular implementation
Readability	5	Clear naming conventions and easy-to-read code
Reusability	5	Components can be reused in
Documentation / Comments	4	Important sections are
Overall Score	4.75 / 5	High-quality, maintainable, and

Conclusion

The Human Development Index (HDI) Prediction Using Machine Learning project follows good software engineering practices by maintaining a clean code structure, meaningful naming conventions, modular design, and reusable components. These practices improve readability, simplify future enhancements, and make the application easier to maintain and extend.